

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15B.1

CARBONYL COMPOUNDS AND THEIR PHYSICAL PROPERTIES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15B.1: Carbonyl Compounds and Their Physical Properties
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	7 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15B.1

Q1. 1. (a) Compare the boiling points of propanal (59°C), propan-1-ol (97°C), and butane (0°C). Explain these differences in terms of intermolecular forces.

[4 Marks]**WCH14/01/Jan23/Q7(a)**

Q2. (b) Explain why lower aldehydes and ketones are soluble in water, but their solubility decreases as chain length increases.

[3 Marks]**WCH14/01/Jan23/Q7(b)**

SECTION B: WORKED MARK SCHEME — 15B.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Compare the boiling points of propanal (59°C), propan-1-ol (97°C), and butane (0°C). Explain these differences in terms of intermolecular forces. 1. Butane has London dispersion forces only (weakest). 2. Propanal has polar C=O bond causing permanent dipole-dipole interactions (stronger than London forces). 3. Propan-1-ol has -OH group forming hydrogen bonds (strongest intermolecular force). 4. Higher energy required to break hydrogen bonds in propan-1-ol -> highest boiling point.	[4]
Q2	Q2: (b) Explain why lower aldehydes and ketones are soluble in water, but their solubility decreases as chain length increases. 1. Lone pair on carbonyl oxygen forms hydrogen bonds with delta+ H of water molecules. 2. As hydrocarbon chain length increases, non-polar hydrophobic alkyl group dominates. 3. Disrupts hydrogen bonding network in water -> solubility decreases.	[3]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15B.1**FAQ #1: Can aldehydes form hydrogen bonds with themselves?****Category: Intermolecular Forces****COMMON EXAMINER TRAP / MISTAKE:**

Claiming aldehydes have hydrogen bonding between aldehyde molecules.

MODEL ANSWER / EXAMINER EXPECTATION:

No! Aldehydes do NOT have a H atom directly bonded to O, so they cannot form H-bonds with other aldehyde molecules (only dipole-dipole). However, they CAN form H-bonds with water because water has O-H.